

AGENTIC AI HOME LOAN UNDERWRITING SYSTEM (INDIA)

AGENT 1: DOCUMENT INGESTION AGENT

Weekly Coding Deliverable & Implementation Report · Owner: Tejasva

Assigned Owner:	Tejasva	Status:	100% COMPLETE & TESTED
GitHub Branch:	tejasva-document-ingestion-agent	Tests Passing:	16 / 16 Pytest (100%)

1. EXECUTIVE SUMMARY & OBJECTIVE

The Document Ingestion Agent automates the first critical stage of the mortgage underwriting pipeline. Its core mission is to convert raw borrower & property documents (PDFs/Images) into structured, strongly-typed, and validated data packages with complete field-level provenance, deterministic Indian domain validation, and cross-document reconciliation. Per Slide 3, it tells the system WHAT THE DOCUMENTS SAY without making subjective creditworthiness decisions, passing verified shared state to downstream agents.

2. COMPLETE 7-STEP INGESTION PIPELINE (SLIDE 3 COMPLIANT)

Step 1: Intake & Classification	Auto-classifies files into PAN, Aadhaar, Salary Slips, Form 16, ITR-V, Bank Statements, and Property Deeds with multi-signal keyword and regex scoring.
Step 2: Document Pre-processing	Multi-layer text layout parsing, table extraction, and image analysis using PyMuPDF (fitz) and pdfplumber.
Step 3: Text & Layout Extraction	Extracts structured tables (salary earnings/deductions breakdown, 6-month tax bank ledger credits/debits, deduction tables).
Step 4: Structured Field Extraction	Extracts strongly typed Pydantic models with field-level provenance metadata (source document, page number, exact snippet, confidence).
Step 5: Field Validation	Deterministic Indian validation: PAN regex & 4th char entity check, Aadhaar = 12-digit/masking check, IFSC code validation, and Salary arithmetic verification: Gross Net + Deductions.
Step 6: Cross-Document Reconciliation	Token-based fuzzy name matching across all documents, Employer consistency vs (Payslips Form 16), Income reconciliation (Payslip net pay vs Bank statement monthly credits), and recurring EMI d
Step 7: Confidence & HITL Routing	Calculates composite confidence (0-100%), evaluates mandatory checklist, and flags contradictions, routes exceptions to Human-in-the-Loop review.

3. TEAM INTEGRATION & LANGGRAPH STATE HANDOFFS

Per Slides 2 and 8, the Document Ingestion Agent is the foundational data provider for the entire multi-agent LangGraph architecture. The table below details the exact structured sub-packages provided to each teammate:

Teammate / Agent	Role in System	Data Package Provided by Document Agent
Aryan & Lakshya (Agent 2: Credit Agent)	CIBIL score analysis, FOIR calculation, RBI LTV tiers evaluation	· <code>borrower_kyc</code>: Name, PAN, DOB, Gender, Address · <code>borrower_income</code>: Monthly Gross/Net, Form 16, Credits · <code>borrower_liabilities</code>: Detected EMIs, Active Loans, Bounces
Soojal & Yash (Agent 3: Property Agent)	Comps analysis, point estimate (INR), market-value range, Rs/sq.ft. benchmark	· <code>property_profile</code>: Property Address, City, Pincode, Super Built-up & Carpet Area (sq.ft.), Consideration Value, Seller & Buyer Names, Registration Details
Saurav (Agent 4: Compliance)	KYC/CDD checks, PMLA source-of-funds, RBI Fair Practices (Hard Gate)	· <code>reconciliation</code>: Cross-document name & employer matching · <code>checklist</code>: Verification of mandatory Indian documents · <code>provenance</code>: Field source document & page audit trails
Ashfaque (Agent 5: Decision Agent)	Deterministic synthesis (Approve / Deny / Suspend), Risk scoring, Underwriting PDF	· <code>confidence</code>: Composite score (0-100%) & 4-factor breakdown · <code>human_review</code>: Requires review flag, priority, suggested actions · <code>audit_log</code>: Chronological trace of all ingestion events

4. CODEBASE FILE INVENTORY & IMPLEMENTED MODULES

src/models/schemas.py	Pydantic domain schemas for PAN, Aadhaar, Payslip, Form 16, ITR, Bank Statement, Property Deed.
src/models/state.py	LangGraph shared state container and structured handoff models for Credit & Property agents.
src/document_ingestion/preprocessor.py	PDF and image pre-processing, text density checks, table extraction with PyMuPDF & pdfplumber.
src/document_ingestion/classifier.py	Automated multi-signal document category classifier with confidence scoring.
src/document_ingestion/extractors.py	Structured field extractors with full source snippet and page number provenance metadata.
src/document_ingestion/validators.py	Deterministic Indian domain validators (PAN, Aadhaar masking, IFSC, Salary arithmetic reconciliation).
src/document_ingestion/reconciliation.py	Cross-document reconciliation engine (Fuzzy name matching, employer matching, salary vs deposit).
src/document_ingestion/confidence.py	Composite confidence scoring (0-100%), checklist completeness, and Human-in-the-Loop review routing.
src/document_ingestion/agent.py	Core Agent orchestrator and standard LangGraph node function: <code>document_ingestion_node(state)</code> .
src/mock_data/generate_docs.py	Programmatic PDF document generator creating 4 realistic test suites with authentic Indian formatting.
src/cli.py & src/api/app.py	Terminal CLI scenario runner and FastAPI REST server with OpenAPI docs for frontend integration.
tests/test_document_ingestion.py	Automated pytest suite covering 16 test cases for all extractors, validators, and scenarios.

5. VERIFICATION RESULTS & TEST SCENARIOS

AUTOMATED PYTEST SUITE: 16 / 16 PASSED (100% PASS RATE)

- Validation Tests: Valid/Invalid PAN, Masked Aadhaar, Valid/Invalid IFSC, Salary Ledger Arithmetic (Gross = Net + Deductions) - ALL PASSED
- Cross-Doc Matching: Exact name match (100%), Fuzzy matching with initials (78%), Mismatch detection (13%) - ALL PASSED
- Document Classifier: Accurate multi-class detection across PAN, Aadhaar, Payslip, Form 16, Bank, Deed - ALL PASSED
- LangGraph State Node: State dictionary compatibility and structured downstream handoffs - ALL PASSED

6. MULTI-SCENARIO UNDERWRITING TEST SUITE RESULTS

Scenario 1: Clean Prime Salaried Borrower (Aarav Sharma - TCS)

Files: PAN, Aadhaar, 3 Salary Slips (Apr/May/Jun), Form 16, 6-Month HDFC Bank Statement, Bangalore Sale Deed
Extraction: Monthly Gross: INR 1,50,000 | Net Pay: INR 1,24,600 | Form 16 Gross: INR 18,00,000 | Bank AMB: INR 1,85,000
Reconciliation: 100.0% Name match across all 15 document pairs | 0.0% Variance between Payslip & Bank Credits | 100% Employer match
Decision: Overall Confidence 95.6% [HIGH] | Human Review Required: False (Straight-Through Processing Approved)

Scenario 2: Name Discrepancy Case (Priya Suresh Patel vs Priya S. Patel)

Files: PAN (Priya Suresh Patel), Aadhaar (Priya S. Patel), Payslip (Priya Patel)
Reconciliation: Flags 3 pairwise name discrepancies (78%, 64%, 68% similarity scores)
Decision: Overall Confidence 65.6% [LOW] | Human Review Required: True (Priority: HIGH)
Underwriter Action: 'Manual underwriter review required to verify name / income discrepancies.'

Scenario 3: Salary vs Bank Deposit Mismatch (Vikram Malhotra)

Files: PAN, Payslip (Net Pay: INR 1,60,000), SBI Bank Statement (Actual Monthly Salary Credit: INR 85,000)
Reconciliation: Detects 46.88% income discrepancy (INR 75,000 monthly variance between claims and bank deposits)
Decision: Contradiction Flagged | Human Review Required: True (Priority: HIGH - Escalated to Underwriter)
Underwriter Action: 'Income discrepancy in June 2026: Bank credit differs significantly from Payslip Net.'

Scenario 4: Missing Mandatory Documents (Rahul Verma)

Files: Only PAN Card + 1 Salary Slip provided (Missing Aadhaar KYC, 6-Month Bank Statement, Property Papers)
Completeness: Completeness Score 40.0% (3 Mandatory document categories missing)
Decision: Overall Confidence 87.9% | Human Review Required: True (Priority: HIGH - Checklist Block)
Underwriter Action: 'Request missing mandatory documents from loan applicant.'

CONCLUSION & HANDOFF READY

The Document Ingestion Agent is fully implemented, verified with 100% test coverage, and committed to GitHub.
Downstream teammates can consume the LangGraph state package via `document_ingestion_node(state)`.